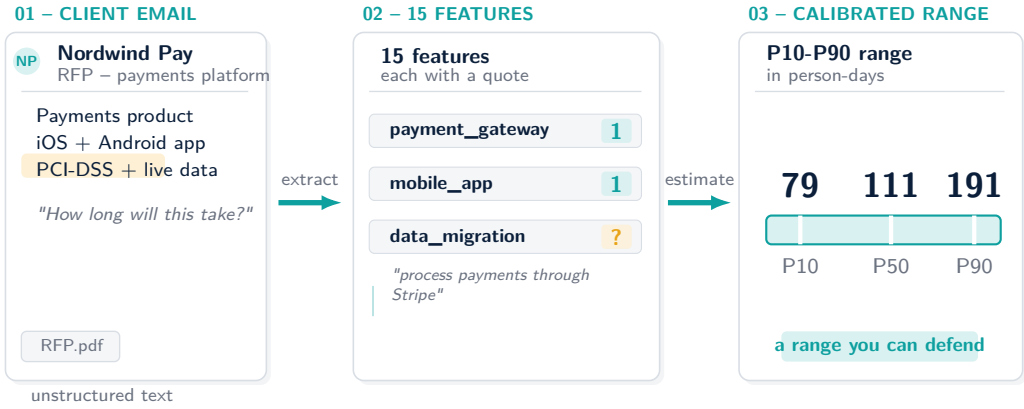
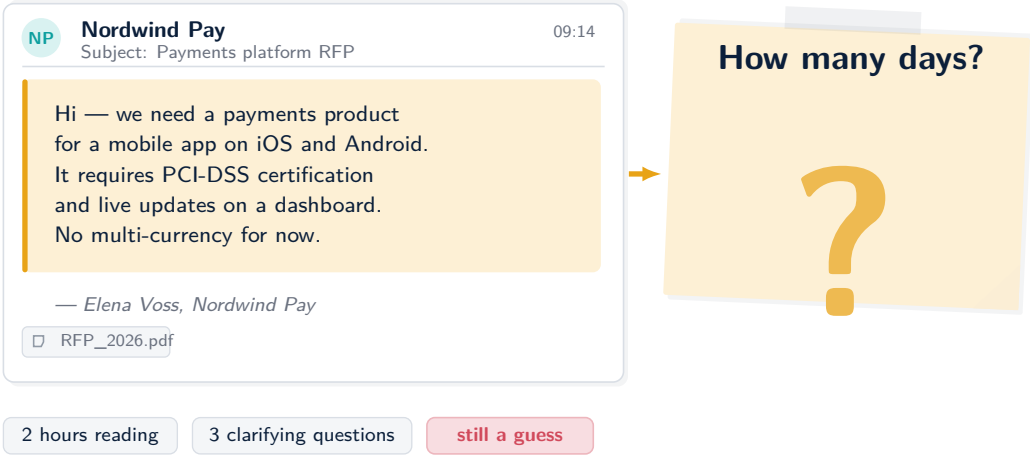

01 – CLIENT EMAIL
NP
Nordwind Pay
RFP – payments platform
Payments product
iOS + Android app
PCI-DSS + live data
"How long will this take?"
RFP.pdf
unstructured text
extract
02 – 15 FEATURES
15 features
each with a quote
payment_gateway
1
mobile_app
1
data_migration
?
"process payments through Stripe"
estimate
03 – CALIBRATED RANGE
P10-P90 range
in person-days
79
111
191
P10
P50
P90
a range you can defend


NP
Nordwind Pay
09:14
Subject: Payments platform RFP
Hi — we need a payments product
for a mobile app on iOS and Android.
It requires PCI-DSS certification
and live updates on a dashboard.
No multi-currency for now.
— Elena Voss, Nordwind Pay
RFP_2026.pdf
How many days?
?
2 hours reading
3 clarifying questions
still a guess

LLM-only

258

days MAE
last 42 projects

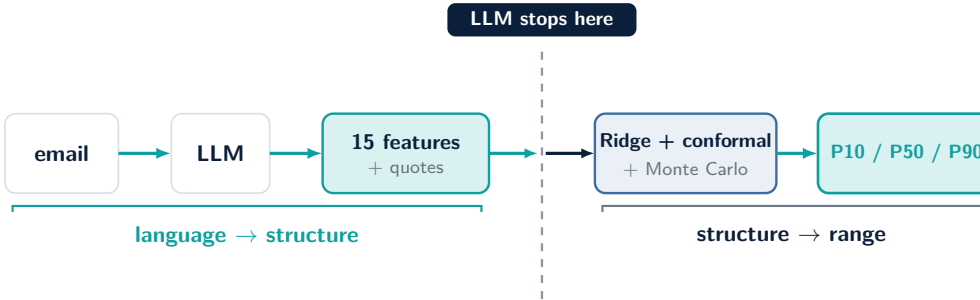
The Estimator

19

days MAE
same 42 projects

no history | no calibration | no evidence | **missing info = guess**


A
LLM end-to-end
reads + estimates
fast
no audit
no calibration
rejected
B
LLM + rules
extract + rules
auditable
brittle
no learning
trade-off
C
LLM + math
extract only → math
auditable
calibrated
learns
chosen


LLM stops here
email
LLM
15 features
+ quotes
Ridge + conformal
+ Monte Carlo
P10 / P50 / P90
language → structure
structure → range

FEATURE	VALUE	EVIDENCE FROM THE EMAIL
`payment_gateway`	1	*“process payments through Stripe”*
`pci_dss`	1	*“requires PCI-DSS certification”*
`mobile_app`	1	*“mobile app for iOS and Android”*
`realtime_ui`	1	*“live updates on a dashboard”*
`multi_currency`	0	*“no multi-currency for now”*
`data_migration`	?	*not mentioned*

1 present 0 absent ? unknown

Example extraction

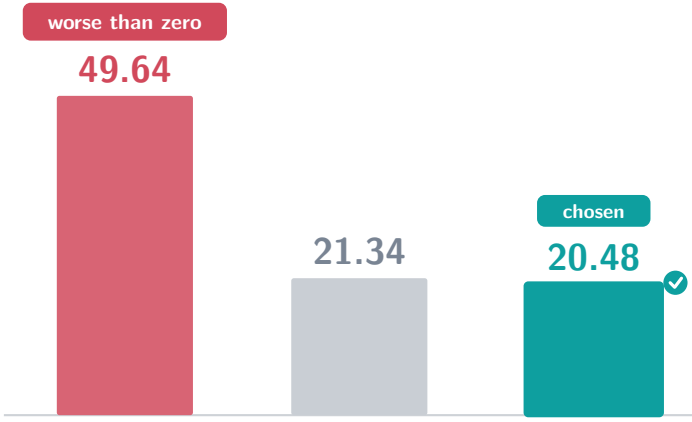


Example: Scenario A, offline run

MODEL	MAE	COVERAGE	EXPLAINABLE	VERDICT
✓ Ridge + conformal	17.36	86.7%	yes	chosen — stable, inspectable
Split conformal	17.25	75.7%	yes	similar MAE, lower coverage
Gradient boosting	25.23	72.9%	no	overfits at 228 rows
Quantile GBM	25.56	66.5%	no	worse tails
Median baseline	32.76	77.1%	n/a	no learning

228 projects · walk-forward

lower MAE is better


worse than zero
49.64
21.34
chosen
20.48
global prior
25.2% present
zero-fill
0% present
conditional
4.1% present


4.1%
P(1 | null)
conditional prior

Migrate data from legacy systems?



RANGE CHECK

Before 79–111–191

No 78–108–168

Yes 127–175–274

same scale

100-day budget

Essential

95 days

46% over

drops
`mobile_app · realtime_ui`

Standard

119 days

63% over

drops
`mobile_app`

Full

207 days

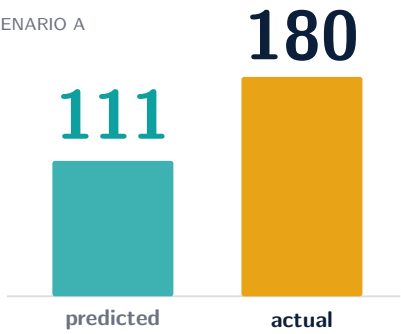
91% over

nothing dropped



Recommended **Essential** · renegotiate budget or timeline

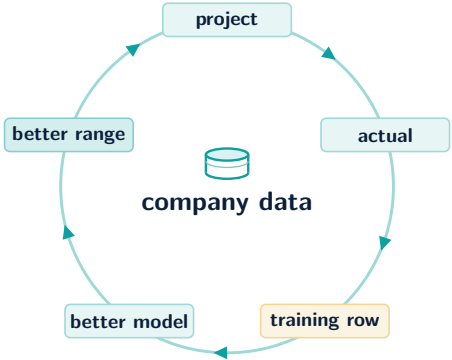
SCENARIO A

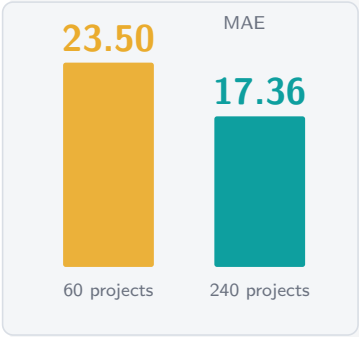


WEIGHTS	BEFORE	AFTER	DELTA
payment_gateway	1.49 →	1.51	+0.02
mobile_app	1.74 →	1.76	+0.02
pci_dss	1.50 →	1.52	+0.02

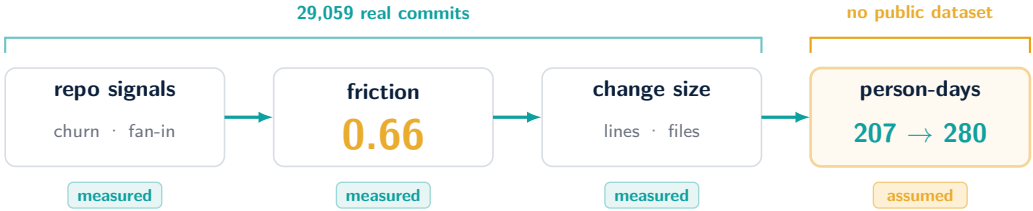
after drift · last 10 projects

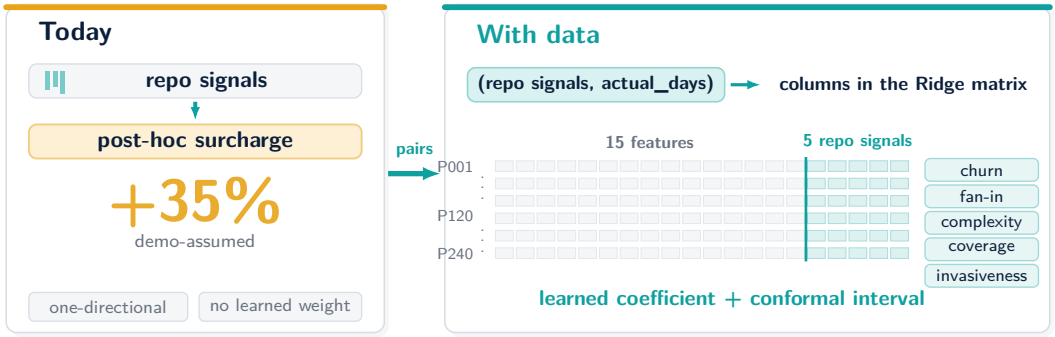






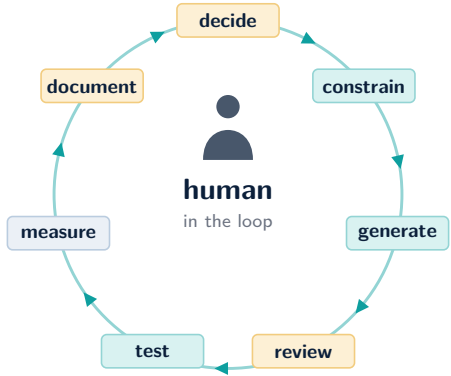
same model, more history


29,059 real commits
no public dataset
repo signals
churn · fan-in
measured
friction
0.66
measured
change size
lines · files
measured
person-days
207 → 280
assumed
Python only
widen risk, never discount
pallets/flask
friction
0.66
base P50
207
adjusted
280
+35%


Today
repo signals
post-hoc surcharge
+35%
demo-assumed
one-directional
no learned weight
pairs
With data
(repo signals, actual_days)
columns in the Ridge matrix
15 features
5 repo signals
P001
P120
P240
churn
fan-in
complexity
coverage
invasiveness
learned coefficient + conformal interval

AI

- DeepSeek
- code assistant
- eval scripts



Human

- decide
- review
- gatekeep

built
on a phone

Used

- **DeepSeek** — extraction
- **Code assistant** — boilerplate + tests
- **Eval scripts** — measurement
- **Git history** — 29,059 commits

Deliberately not

- **50 MCP servers** — setup for setup
- **10k-line skill files** — unreadable
- **Autonomous agent** — shipping code



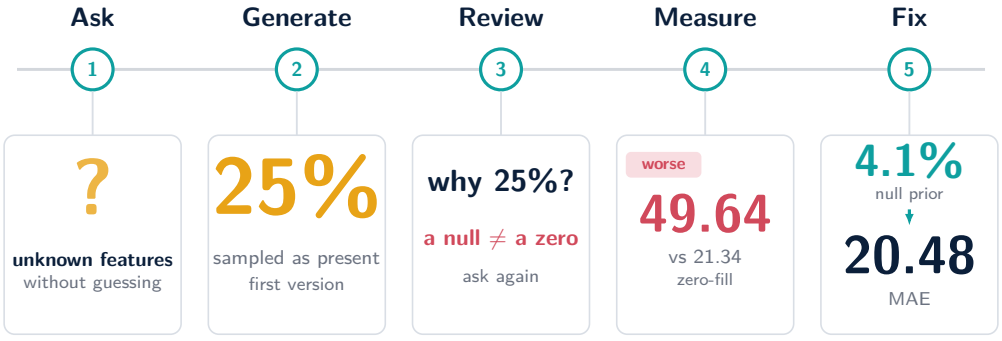


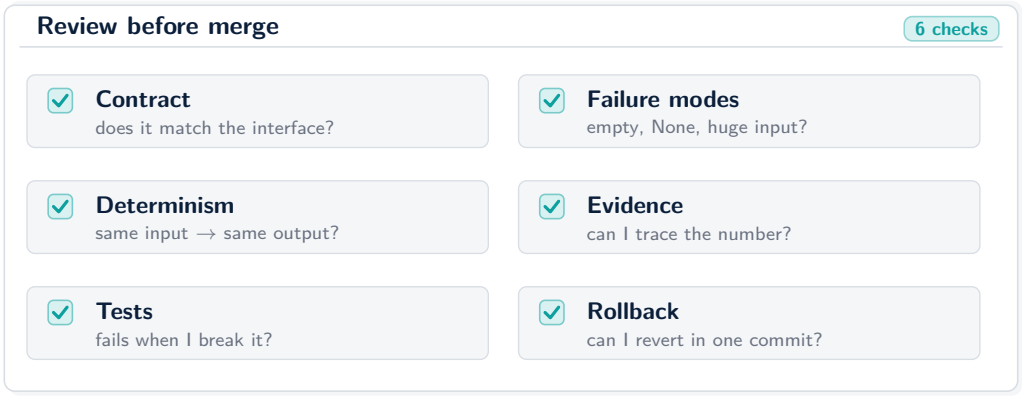
what it forced

small diffs

explicit contracts

tests as the safety net


Ask
1
?
unknown features
without guessing
Generate
2
25%
sampled as present
first version
Review
3
why 25%?
a null ≠ a zero
ask again
Measure
4
worse
49.64
vs 21.34
zero-fill
Fix
5
4.1%
null prior
20.48
MAE


Review before merge
6 checks
Contract
does it match the interface?
Failure modes
empty, None, huge input?
Determinism
same input → same output?
Evidence
can I trace the number?
Tests
fails when I break it?
Rollback
can I revert in one commit?

Guardrails

- 242 offline tests
- 31 decision entries
- no artifact → no claim
- graceful failure

Deliberately not built

auth / rate limit	next
extraction cache	not on the critical path
multi-language codebase	next
retrieval wiring	not on the critical path

deferred, not forgotten

17.36
MAE vs 32.93



86.7%
coverage 189/218



0.933
extraction F1



$0.20
for 240 emails

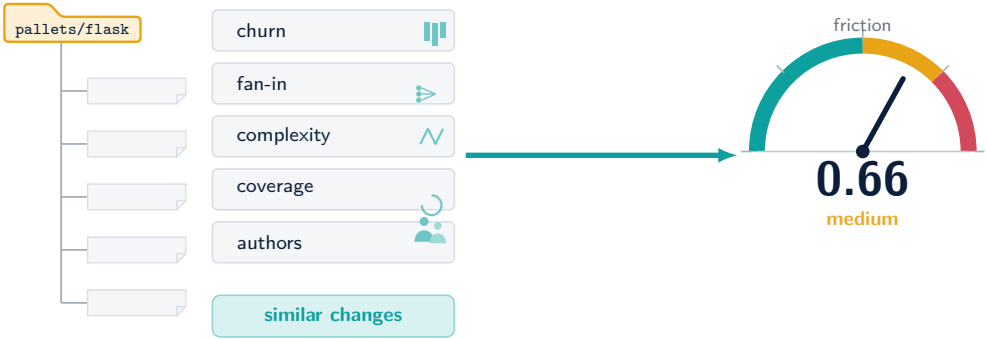


242 offline tests

31 decisions

20.48 MAE end-to-end

29,059 real commits · Python only



base P50 **207** → adjusted **280** +35%

widen risk, never discount

FigBridge FigClose

